

MAURICE BARAZA OTIENO

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EDUCATION

Lafayette College

Easton, PA

Bachelor of Science Electrical and Computer Engineering

Graduation: May, 2026

Relevant Coursework: Embedded Systems, Digital circuits and computer Architecture, Signals and Communication Systems, Data Structures and Algorithms, Mixed Signal-Design of electronic circuits, Industrial electronics and control systems, AI for computer vision, Cybersecurity

Relevant Skills: Embedded design with Arduino, STM32, RASP BERRY PI, FPGA validation (System Verilog, Xilinx Vivado), circuit simulation (LTSpice), PCB design and hardware bring-up (KiCad), Bench testing, C/C++, VS Code, Assembly language (MIPS), Python & MATLAB, java, I2C, UART, SPI, Signal Integrity. Timing circuits

Awards and Honors: Murray G. Clay '30 Award Fund in Engineering Science, Dean's List

Campus Engagement: IEEE, Lafayette African and Caribbean Students Association, Soccer club and National Society of Black Engineers.

RELEVANT EXPERIENCE

Kenya Power and Lighting Company: Junior Intern, Migori Kenya

July, 2025 - August, 2025

- Read and recorded electricity consumption for more than 50 residential and commercial meters, supporting load monitoring and billing accuracy
- Participated in preventive maintenance and inspection of distribution transformers to ensure system reliability and safety compliance.
- Assisted in replacement of faulty transformer connection components under supervision.

Project Experience

COMPUTER ARCHITECTURE

27 January - 3 March 2024

- Described a basic MIPS Instruction set architecture based pipelined processor with ALU, control registers, Data Paths etc. in System Verilog.
- Verified functional correctness and timing behavior through waveform simulation and testbench development in Vivado

STRIDE – STAIR-CLIMBING MOBILITY ASSIST SYSTEM

20 August 2025 - Present

- Designed the power management system incorporating 24V battery pack, current monitoring, and protection circuitry for safe motor operation. □
- Designed embedded control architecture integrating STM32 microcontrollers, BLDC motor drivers, and sensor feedback for closed-loop mobility control.
- Collaborated with seven teammates to performed hardware integration, PCB design, and system-level debugging across analog, digital, and power subsystems.

DIGITAL DESIGN: PULSE SENSOR AND REACTION TIMER

01 June - 20 August 2022

- Collaborated with a partner to Design and implemented a system Verilog hardware module for pulse detection and reaction-time measurement on a Nexys A7100T FPGA.

- Verified functional correctness and timing behavior through waveform simulation and testbench development in Vivado
- Integrated and debugged the design on hardware via PMOD interfaces, resolving timing and synchronization issues.

https://github.com/otienomaurice/Pulse_Sensor_and_reaction_timer.git

Digital Design: Led Matrix controller

01 June - 20 August 2023

- Designed a system Verilog-based display controller implementing frame buffer storage using on-chip BRAM to drive Adafruit's 32 x 16 LED matrix display panels.
- Verified timing and refresh behavior through simulation waveform analysis in Vivado.

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

Equity Group Foundation: Intern, College Counselling

- Mentored recent high school graduates on academic planning, university applications, and career development.
- Provided structured guidance sessions focused on goal setting, scholarship preparation, and transition to tertiary education.

ADDITIONAL EXPERIENCE

ECE Research fellow, Lafayette college

06/26/ 2025-present

Student IT Assistant, Tech Lounge, Lafayette College

01/26/ 2024-present

Student Library Assistant, Lafayette College

08/22/ 2023-present

Additional Skills

Languages: English (Fluent) Swahili (Fluent), Luo (Fluent), Sheng (Conversational)